"Banana Leaf Spot Disease Classification"

21CSC406J/ Neural Networks and Deep Learning

CBL REPORT

Submitted by

Kamisetty Koushik RA2311026050059

Under the guidance of

Dr. R. Venkatesh

Associate Professor, Department of AIML

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GitHub notebook repository: https://github.com/koushik5636N/DL-CBL

Problem and objective

This project classifies a smartphone image of a banana leaf into Healthy, Sigatoka, Cordana or Pestalotiopsis. The intended role is preliminary disease screening, not a replacement for plant-pathology assessment. The experimental objective is to build a reproducible CNN pipeline that controls duplicate-image leakage, handles class imbalance explicitly and assesses performance on a held-out test split.

Macro F1 is the primary selection metric because every class should contribute equally despite the unequal sample counts. Accuracy, per-class precision/recall/F1 and the confusion matrix provide complementary evidence. Three candidates were declared: a scratch CNN, an unweighted ImageNet-pretrained MobileNetV2 and the same transfer model with balanced class weights.

Principal experimental result

The selected unweighted MobileNetV2 classified 171 of 179 test images correctly: 95.53% test accuracy and 94.73% macro F1. Validation macro F1 was 97.14%. Gains over the scratch CNN were 16.76 percentage points in accuracy and 26.10 points in macro F1. All eight errors were retained for analysis.

Evidence: test_results-2.csv; validation_comparison-3.csv; held-out confusion matrix reproduced from the executed notebook.

Dataset and integrity audit

BananaLSD version 1 provides 937 original 224 × 224 RGB images and a separate pre-augmented set. Images were collected with three smartphones in Gazipur, Bangladesh, during June 2021 and labelled by a plant pathologist. Only OriginalSet was used; the supplied AugmentedSet was excluded. Dataset DOI: 10.17632/9tb7k297ff.1; dataset paper: https://doi.org/10.1016/j.dib.2023.109608.

Table 1. Original and audited class counts; audit totals are from the executed notebook.

Class	Original	After audit	Removed/quarantined
Healthy	129	90	39
Sigatoka	473	473	0
Cordana	162	162	0
Pestalotiopsis	173	172	1
Total	937	897	40

All 937 images were readable. The audit quarantined two exact-content images with conflicting labels and removed 38 repeated images. The retained set therefore contained 897 images. After auditing, Sigatoka represented 52.73% and Healthy 10.03% of the retained data, making imbalance an important experimental consideration.

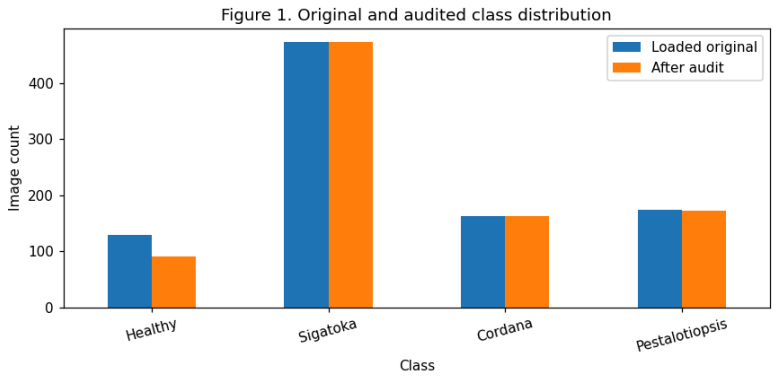


Figure 1. Original versus audited class distribution. Source: saved notebook output.

Dataset splits and preprocessing

A 64-bit difference hash (dHash) with Hamming distance at most 4 identified six visually similar pairs. Connected components yielded 891 similarity groups across 897 audited images. Similar images were grouped rather than deleted. StratifiedGroupKFold supplied the first outer five-fold split for test allocation and the first inner four-fold split for validation, using seeds 42 and 43. This creates one approximately 60/20/20 holdout, not cross-validated performance estimates.

Assertions confirmed that no relative path, decoded-pixel hash or detected-similarity group appeared in more than one split. Perceptual hashing is only a heuristic: the absence of detected overlap does not establish plant-, leaf- or field-level independence because those identifiers were unavailable. Grouped splitting follows the non-overlapping-group principle documented by scikit-learn: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedGroupKFold.html.

Table 2. Class-wise split sizes after auditing; all classes are represented in all splits.

Class	Training	Validation	Test	Total
Healthy	55	17	18	90
Sigatoka	282	97	94	473
Cordana	98	32	32	162
Pestalotiopsis	103	34	35	172
Total	538	180	179	897

Pillow applied EXIF orientation correction, RGB conversion and bilinear resizing to 224 × 224. The scratch CNN scales pixels to [0, 1]; MobileNetV2 scales them to [-1, 1]. Horizontal flip, rotation of up to 0.06 turns, zoom of 10% and contrast variation of 10% operate during training only. Validation and test images are not randomly augmented.



Figure 2. Three saved training examples per class. No test samples were used for this EDA display.

Evidence: Notebook audit, split assertions and input pipeline; raw_image_manifest, class_counts and split_manifest exports referenced by the notebook.

Methodology and model design

The scratch CNN establishes a task-specific baseline with three 3 × 3 convolution/ReLU blocks containing 32, 64 and 128 filters; each block is followed by max pooling. Global average pooling, dropout and a four-unit softmax classifier produce class probabilities. The model has 93,764 parameters. The common data split and augmentation configuration allow a controlled comparison with transfer learning.

Both transfer candidates use MobileNetV2 without the ImageNet classification head, followed by global average pooling, dropout and a four-class softmax head. Each has 2,263,108 parameters. The backbone is frozen during head training; the final 30 backbone layers are then eligible for fine-tuning except BatchNormalization layers. The backbone runs with training=False to preserve batch-normalization statistics. This two-stage design follows Keras transfer-learning guidance: https://keras.io/guides/transfer_learning/.

Table 3. Hyperparameters recorded in the uploaded notebook.

Setting	Implemented value
Input / batch size	224 × 224 × 3 RGB / 32
Random seed	42; inner split seed 43
Optimizer / loss	Adam / sparse categorical cross-entropy
Initial learning rate	0.001 for scratch and frozen-head training
Fine-tuning learning rate	0.00001
Maximum epochs	Scratch: 20; transfer head: 12; fine-tuning: 12
Early stopping	Validation macro F1; patience 4; mode=max
Checkpoint rule	Best validation macro F1 in each phase
Learning-rate schedule	ReduceLROnPlateau: factor 0.5, patience 2, floor 0.0000001
Regularisation	Global average pooling and dropout 0.30
Training augmentation	Horizontal flip, rotation 0.06 turns, zoom 0.10, contrast 0.10
Fine-tuning scope	Last 30 backbone layers, excluding BatchNormalization
Model selection	Validation macro F1, then accuracy, then name

Class-imbalance intervention

The weighted MobileNet candidate changes only the loss weighting, using class frequencies from the 538 training images. Balanced weights follow n_training / (4 × class_training_count), so minority-class errors receive greater loss weight. Validation and test metrics remain unweighted. Method documentation: https://scikit-learn.org/stable/modules/generated/sklearn.utils.class_weight.compute_class_weight.html.

Table 4. Training-only class weights for the weighted MobileNet candidate.

Class	Training images	Balanced weight
Healthy	55	2.4455
Sigatoka	282	0.4770
Cordana	98	1.3724
Pestalotiopsis	103	1.3058

The design goes beyond one baseline through transfer learning, a controlled class-weight comparison, duplicate-content auditing, group-based splitting and Grad-CAM error inspection. Each component is evaluated as evidence from this run; none establishes a universally superior model or deployment readiness.

Evidence: Notebook configuration, model builders and training utilities; parameter counts and completed phase durations in validation_comparison-3.csv.

Validation and learning behaviour

Table 5. Validation results only; these values were used for checkpoint and model selection.

Candidate	Val. accuracy (%)	Val. macro F1 (%)	Epochs run	Parameters
Scratch CNN	81.67	75.92	20	93,764
MobileNetV2, unweighted	97.22	97.14	17	2,263,108
MobileNetV2, weighted	97.22	96.88	20	2,263,108

Unweighted MobileNetV2 was selected before the test set was opened because its validation macro F1 was highest. Its margin over the weighted variant was 0.26 percentage points; equal validation accuracy did not override this criterion. The pre-test selection JSON records the chosen model, dataset fingerprint and model hashes.

Figure 3. Learning curves; checkpoints selected using validation only

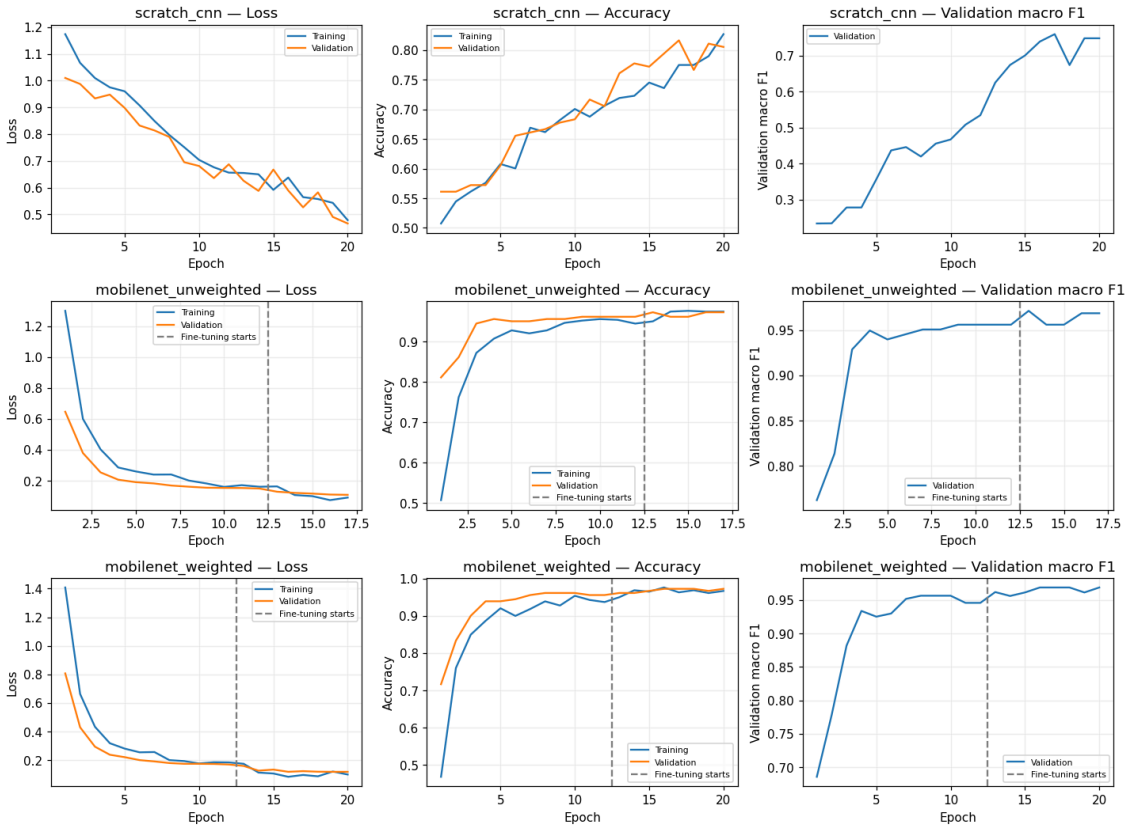


Figure 3. Saved training/validation loss, accuracy and validation macro-F1 curves for all candidates. Dashed lines mark the start of fine-tuning.

The scratch CNN improved throughout training: validation loss decreased from approximately 1.01 to 0.47, while training and validation accuracy stayed broadly similar. Its best validation macro F1 was 75.92%, below the transfer models. This is more consistent with limited representation or incomplete convergence under the specified budget than with severe late-stage overfitting; it does not prove that a longer run would improve held-out performance.

The unweighted transfer model completed 12 head-training epochs and five fine-tuning epochs. The best fine-tuning macro F1 occurred at the first fine-tuning epoch; subsequent epochs did not surpass it, triggering early stopping. The weighted model completed 12 head epochs and eight fine-tuning epochs. Both achieved high, broadly stable validation performance without a large sustained training–validation accuracy gap. Training augmentation and weighted loss mean the plotted losses are not directly equivalent to unaugmented validation loss.

Evidence: validation_comparison-3.csv; selection_before_test-4.json; executed epoch logs and Figure 3. The class-weight result is one seed on one holdout, not a statistical significance claim.

Held-out test results

After locking the model selection, the notebook evaluated the three predeclared candidates in one final test round on the same 179 images. The selected model remains unweighted MobileNetV2. The weighted candidate has slightly higher observed test scores, but this does not justify changing the final model after test exposure.

Table 6. Test results for predeclared candidates; time is observed training wall time, not inference latency.

Candidate	Selected	Test acc. (%)	Test macro F1 (%)	Time (s)
Scratch CNN	No	78.77	68.64	58.17
MobileNetV2, unweighted	Yes	95.53	94.73	63.03
MobileNetV2, weighted	No	96.09	95.61	68.11

The weighted model exceeded the selected model by 0.56 percentage points of test accuracy and 0.88 points of test macro F1. These are descriptive observations only. The majority-class test reference is 52.51% accuracy (predicting Sigatoka for all 179 images). The selected model achieved 93.77% balanced accuracy, calculated as mean class recall.

Table 7. Selected unweighted MobileNetV2 per-class test performance. All metric entries are percentages.

Class	Precision (%)	Recall (%)	F1 (%)	Test images
Healthy	94.12	88.89	91.43	18
Sigatoka	94.85	97.87	96.34	94
Cordana	100.00	96.88	98.41	32
Pestalotiopsis	94.12	91.43	92.75	35

Figure 4. Held-out test confusion matrix — mobilenet_unweighted

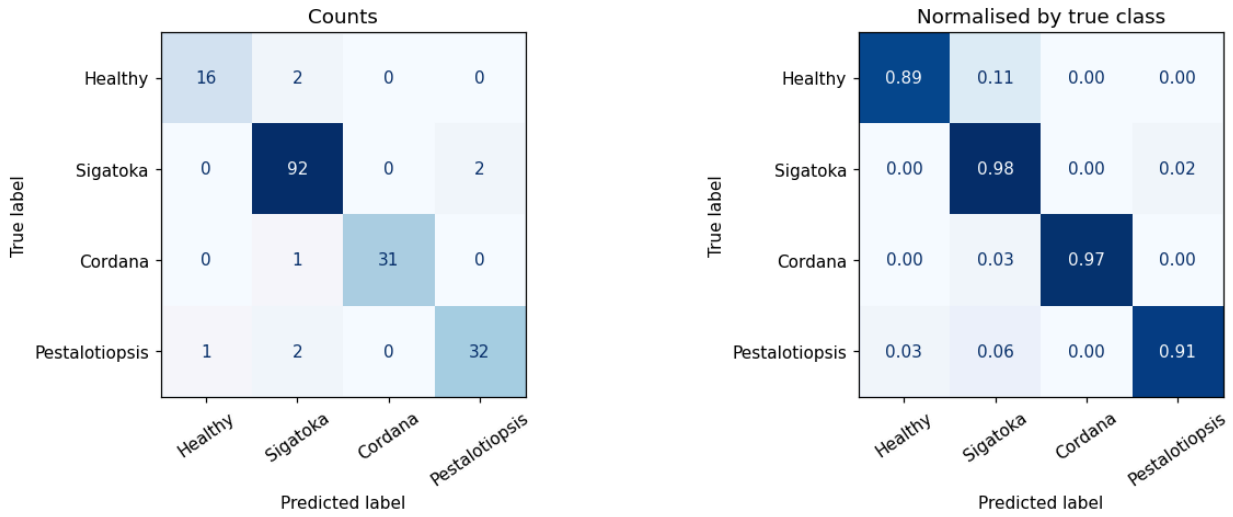


Figure 4. Selected-model test confusion matrix: counts and row-normalised proportions; rows are true classes.

There are eight errors: Healthy → Sigatoka (2), Sigatoka → Pestalotiopsis (2), Cordana → Sigatoka (1), Pestalotiopsis → Healthy (1) and Pestalotiopsis → Sigatoka (2). Three directions tie for the largest count; there is no unique dominant error pair. Healthy has the lowest recall at 88.89% (16/18), whereas Sigatoka recall is 97.87% (92/94). Cordana precision is 100.00% on this split, not a guarantee of zero future false positives.

Evidence: test_results-2.csv; Figure 4 and notebook per-class report. Reported per-class values were independently recomputed from the displayed count matrix and reproduce macro F1 94.73%.

Misclassified-image analysis

The diagnostic panel shows the four most confident selected-model errors, not four arbitrarily chosen examples. Confidence is the largest softmax output for an incorrect prediction; it is not the measured probability that the diagnosis is clinically correct. The following explanations are visual hypotheses from the supplied images, not verified biological causes.

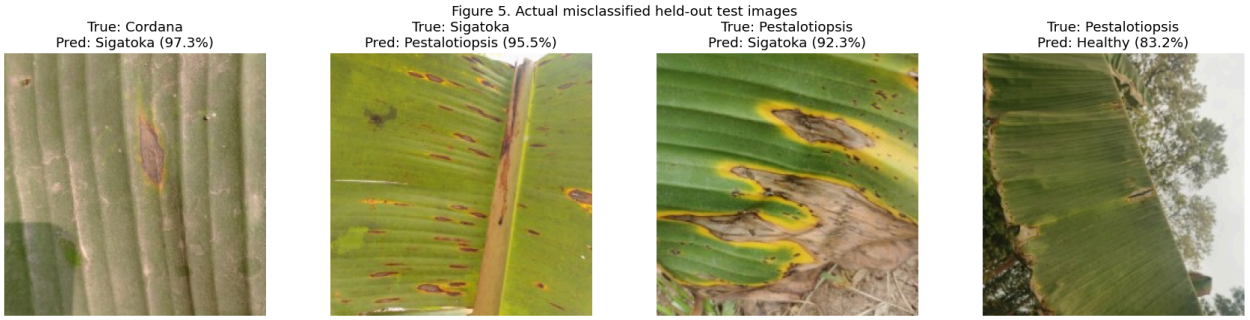


Figure 5. Four actual held-out errors from the uploaded run; labels and confidence are copied from saved predictions.

Table 8. All eight selected-model test errors, ordered by saved confidence.

Relative image path	True class → prediction	Confidence (%)
Cordana/140.jpeg	Cordana → Sigatoka	97.29
Sigatoka/288.jpeg	Sigatoka → Pestalotiopsis	95.47
Pestalotiopsis/64.jpeg	Pestalotiopsis → Sigatoka	92.27
Pestalotiopsis/75.jpeg	Pestalotiopsis → Healthy	83.17
Healthy/119.jpeg	Healthy → Sigatoka	79.92
Healthy/37.jpeg	Healthy → Sigatoka	63.43
Pestalotiopsis/108.jpeg	Pestalotiopsis → Sigatoka	47.86
Sigatoka/277.jpeg	Sigatoka → Pestalotiopsis	39.76

Four image-specific explanations

Cordana/140.jpeg — Cordana → Sigatoka (97.29%). The image contains a single prominent elongated brown lesion with yellow edging on a muted green leaf. Similar colour and shape cues may overlap with patterns learned for Sigatoka. Grad-CAM includes the lesion and surrounding central leaf tissue, so attention to a lesion-like region did not ensure the correct class.

Sigatoka/288.jpeg — Sigatoka → Pestalotiopsis (95.47%). Numerous brown elongated spots and yellow discolouration occur near the midrib. The broad central activation overlaps several lesions and the midrib rather than isolating one distinguishing feature. Shared necrotic appearance is a plausible source of confusion; the image does not establish the exact learned decision rule.

Pestalotiopsis/64.jpeg — Pestalotiopsis → Sigatoka (92.27%). Large tan necrotic patches with bright yellow margins are visible, including damaged tissue near the leaf edge. The heatmap emphasises part of this affected tissue. Colour and lesion-pattern overlap may contribute to the wrong class even when the activation lies on visibly damaged leaf regions.

Pestalotiopsis/75.jpeg — Pestalotiopsis → Healthy (83.17%). Much of the photograph is a green whole leaf, while visibly damaged areas occupy a relatively small part of the image. Activation is concentrated mainly on broad leaf-surface regions. Dilution of small lesion evidence at 224 × 224 is a plausible explanation for the healthy prediction, not a demonstrated causal mechanism.

A further low-confidence Sigatoka error (277.jpeg, 39.76%) was automatically flagged for low edge variance relative to the training-set tenth percentile. This suggests inspecting blur or a feature-poor crop, but no definitive visual explanation is assigned because that image is not displayed in the submitted four-example panel.

Evidence: misclassification_analysis-6.csv; Figure 5; Figure 6. Automated brightness/sharpness flags are inspection prompts, not validated diagnoses.

Grad-CAM interpretation

Grad-CAM was computed for the predicted-class logit using the final MobileNetV2 feature maps. Spatially averaged score gradients weight feature channels; positive activations are normalised and overlaid on the original image. The notebook bypasses random augmentation during this diagnostic. Method attribution: Selvaraju et al., ICCV 2017, DOI 10.1109/ICCV.2017.74; Keras implementation guidance: https://keras.io/examples/vision/grad_cam/.

Figure 6. Selected-model Grad-CAM diagnostics

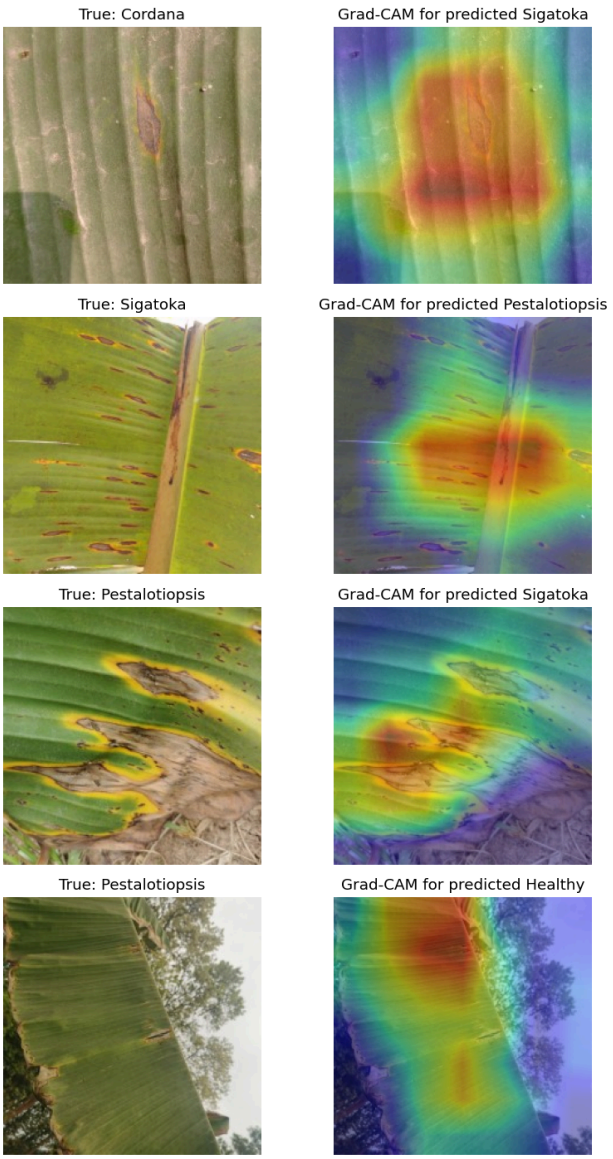


Figure 6. Predicted-class Grad-CAM overlays for the same four held-out errors, in Figure 5 order.

The first three examples show substantial activation on or near visibly affected leaf areas, although the activation is broad and sometimes includes the midrib or adjacent normal tissue. The fourth example shows strong response on predominantly green leaf surface and weaker emphasis on small damaged areas. Thus, wrong labels are not uniformly explained by attention to the external background; class ambiguity and incomplete lesion coverage remain plausible alternatives.

These maps are coarse diagnostic visualisations, not lesion segmentation masks or causal proof of model understanding. Their colours are normalised within each image and should not be interpreted as comparable calibrated confidence. No lesion annotations, localisation metric, occlusion study or explanation-faithfulness test was available. Any stronger claim about disease-specific reasoning would require additional controlled evidence.

Relevance, limitations and ethics

A possible application is a smartphone-assisted leaf-screening workflow that suggests one of the four known classes for expert review. A sensible interface would request a close, well-lit leaf photograph, show uncertainty and allow abstention or referral. These interface safeguards are recommendations, not features demonstrated by the present notebook. MobileNetV2 is a candidate for resource-constrained use, but phone latency, memory, battery consumption and field usability were not measured.

Generalisability is limited by collection from one geographic area and period, a small audited dataset, one random seed and one holdout. The Healthy test class contains only 18 images. No external field test, repeated-seed uncertainty estimate, confidence calibration, out-of-distribution test or robustness stress test was performed. Similarity grouping reduces detected duplicate leakage but does not replace true plant/field-level separation.

The model is closed-set: an unfamiliar disease, mixed infection, non-leaf photograph or out-of-focus image can still receive a confident known-class label. The Pestalotiopsis → Healthy error is particularly important because it could delay escalation. Conversely, false disease labels could prompt unnecessary treatment. Predictions should not be treated as pesticide prescriptions or a substitute for expert inspection.

Potential bias includes crop variety, symptom severity, camera, lighting and background correlations. Farm photographs may expose people, geolocation or identifying metadata, so a deployed service should minimise collection, strip unnecessary metadata and obtain consent. The dataset is distributed under CC BY 4.0; author and source attribution must be preserved. Dataset terms: https://data.mendeley.com/datasets/9tb7k297ff/1.

Reproducibility and research integrity

The recorded run used Python 3.13.15 and TensorFlow 2.20.0, with a GPU device present. The uploaded output does not establish the GPU model. The pipeline sets Python/NumPy/TensorFlow seeds, requests deterministic TensorFlow operations, stores manifests and model hashes, records package versions and reuses unchanged completed results. Cross-hardware or cross-version numerical equality is not guaranteed. Run identifier: 930c81166314.

The uploaded submission_checks-5.csv passes class representation, decoded-pixel separation, detected-similarity-group separation, pre-test selection locking and completion of one final evaluation round. It still marks error-explanation review, learning-curve discussion and Grad-CAM discussion as incomplete. This report supplies evidence-based discussion but does not change those recorded notebook checks or claim that the student has personally verified them. They must be reviewed and updated honestly before submission.

AI-use disclosure: An AI assistant was used to assist notebook preparation, inspect the provided outputs and draft this report. Numerical results come from the uploaded executed notebook and exported files; the report does not claim a new training run. Visual error explanations are interpretive hypotheses and require student review. The student remains responsible for understanding, checking and defending every reported result.

Conclusion and future work

Validation-selected unweighted MobileNetV2 achieved 95.53% accuracy and 94.73% macro F1 on 179 held-out BananaLSD images, substantially exceeding the scratch CNN under the recorded budget. The next priority is an independently collected, plant/field-separated external test with expert-reviewed error cases; repeated-seed evaluation and calibrated referral thresholds should follow before any deployment claim.

Evidence: Executed notebook; test_results-2.csv; selection_before_test-4.json; submission_checks-5.csv; audit/split outputs. External sources are cited inline where used.

Dataset citation: Arman, S. E., et al. (2023), Banana Leaf Spot Diseases (BananaLSD), Mendeley Data, v1, DOI 10.17632/9tb7k297ff.1. Dataset paper: Arman et al., Data in Brief 50, 109608, DOI 10.1016/j.dib.2023.109608.